Our store has a problem where the relationship between purchase probability and price is completely unknown. Attempting to determine this relationship can be both difficult and expensive because testing different prices will likely have a negative impact on revenue. Different prices of course will yield different profits per customer, with there being several peaks in profit. The average profit per customer at the optimal price point is $11.925, however the price point is unknown so our goal is to find a price point that will get as close as possible to $11.925 average profit per customer.

For deciding my optimal price, I used gradient ascent which treats revenue per customer as a curve and repeatedly moves the price in the direction where that curve rises. Because we don't know the true curve, each step estimates its slope from two experiments: one at a price $1 below the current price and one at a price $1 above it, with both using around 100 customers. The slope is the difference in mean revenue per customer between the two, divided by the gap between the prices. The price then moves by the slope multiplied by a learning rate, so a positive slope raises the price and a negative slope lowers it. The learning rate shrinks with each iteration and every move has a limit of two dollars so that some outlier slope doesn't have too great of an effect on the price. I first ran exploratory searches from seven starting prices which were 10, 25, 35, 45, 50, 60, and 65. Each used 30 iterations and about 200 customers per arm each. Those runs ended at about 12, 27, 28, 46, 52, 54, and 63. The first three landed on the two peaks near $12 and $28, while the rest finished at prices earning only about $1 to $8 in profit per customer. For the final design I narrowed down my starting prices and picked four of the end points from the last set of prices, those being 10, 27, 40, and 55, with 10 iterations per search and about 2,000 customers for each iteration.

To handle uncertainty, I treated every price comparison as a two-sample test on mean revenue per customer. Each customer either pays the full price or nothing, so revenue per customer can have a significant amount of variance. After the searches ended, I tested each finalist on 1,000 new customers and computed its mean revenue per customer and standard error. I then compared the best finalist with the runner-up using a 90% confidence interval for the difference in means, built from the standard error of the difference. I used 90% confidence instead of 95% or 99% because each extra step of confidence requires substantially more customers, and customers tested at a suboptimal price are costly; with 90%, the risk of picking the wrong price is higher, but the expected loss is small when the candidates earn similar revenue. In the reported run, the top two prices of 27.63 and 12.37 differed by 0.895 per customer with a 90% interval of 0.154 to 1.637, so the test stopped after the first round.

One of the main challenges was that the two best peaks, near $12 and $28, earn similar revenue, so separating them takes many customers, especially because revenue per customer is noisier at higher prices where fewer people buy. In one early run, four rounds of extra testing left the two prices statistically tied, and in other runs the same comparison came out clearly in favor of the higher price.

A smaller difficulty was reproducibility; because my experiments all drew from one random number generator, results changed depending on which earlier cells had run. I fixed this by

reseeding the generator immediately before the final search, so the reported numbers are identical on every run.

For the results, my final algorithm chose a price of about $27.63. The search and comparison stages experimented on 11,982 customers in total, earning roughly $109,900 in revenue during those experiments, or an average of $9.17 per customer, which isn't as close to the profit supremum as I'd hoped to be. The main issue was simply the experimentation. As stated previously, the more testing done with various prices, the less profit the store gets. The searches that started at 40 and 55 spent thousands of customers at prices earning only $6 to $8 each, while the comparison rounds at prices near $12 and $28 earned close to the optimum. These figures do not include the seven exploratory searches described earlier, which used about 84,000 additional customers and were used to understand the shape of the revenue curve and choose the final starting prices, so the full cost of developing this design was much higher than the cost of running the final algorithm. In a comparison between the top two finalists, the chosen price beat the runner-up (about $12.37) by 0.895 per customer, with a 90% interval of 0.154 to 1.637, which excluded zero so it was significant. A separate check with 5,000 customers per price estimated the gap at about 0.43, so the chosen price is probably better by a smaller margin than that first comparison suggested. If I applied the chosen price to future customers instead of continuing to experiment, I would expect a better average that could be close to the optimal profit per customer.